



Dual emotion based fake news detection: A deep attention-weight update approach

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ABSTRACT

The proliferation of false information is a growing problem in today's dynamic online environment. This phenomenon requires automated detection of fake news to reduce its harmful effect on society. Even though various methods are used to detect fake news, most methods only consider data-oriented text features; ignoring dual emotion features (publisher emotions and social emotions) and thus lack higher levels of accuracy. This study addresses this issue by utilizing dual emotion features to detect fake news. The study proposes a Deep Normalized Attention-based mechanism for enriched extraction of dual emotion features and an Adaptive Genetic Weight Update-Random Forest (AGWu-RF) for classification. First, the deep normalized attention-based mechanism incorporates BiGRU, which improves feature value by extracting long-range context information to eliminate gradient explosion issues. The genetic weight for the model is adjusted to RF and updated to achieve optimized hyper parameter values that support the classifiers' detection accuracy. The proposed model outperforms baseline methods on standard benchmark metrics in three real-world datasets. It outperforms state-of-the-art approaches by 5%, 11%, and 14% in terms of accuracy, highlighting the significance of dual emotion capabilities and optimizations in improving fake news detection.

1. Introduction

Even though social media makes it easier to share information, many news posts that are shared online are not reliable or of low quality. People face a challenge when it comes to distinguishing between real and fake news on social media given they lack capacity for editorial judgment, third-party filtering, or fact-checking (Allcott & Gentzkow, 2017). Thus, the false information on these podiums misleads readers, which eventually hurts social media's credibility and reliability as a source of news. The most recent examples of fake news were the conspiracy theories about "cures" for COVID-19; one such misinformation involved enticing people to eat soap.¹ Such kinds of false information prey on gullibility of people who believe biased information thus fulfilling the intention of the manipulator. Manually checking facts is time-consuming; as a result, automatic detection of fake news has risen in popularity. Natural Language Processing (NLP) and Artificial Intelligence (AI) are increasingly applied to check facts automatically and save time (de Souza et al., 2020). To prevent the spread of false news online and preserve a healthy internet, the scientific community has increased its efforts to create tools to detect false news. However, it is still difficult to quickly spot fake news (Li, Li, Liu, & Wang, 2022a; Li, Wei, Zhou, Li, &

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¹ <https://www.bbc.com/news/stories-52731624>

Jin, 2022c) since information on social media tends to be a mix of true and false information intended to trick and manipulate the reader's emotion.

There have been numerous recent efforts and attempts to address the problem of fake news on social media (Sahoo & Gupta, 2021). Some studies use implicit emotional cues (Giachanou et al., 2021), sentiment analysis (Ajao et al., 2019), textual (Popat et al., 2016), credibility cues (Castillo et al., 2011), and others adopt social engagement (Shu et al., 2018; Shu et al., 2019b; Shu et al., 2019a). These techniques only consider the semantics and, in some cases, the emotions evoked by news content and ignore user comments, even though emotions are a crucial factor in people's vulnerability to misperceptions and obfuscation of the real-world (Martel et al., 2020; Li et al., 2022a). Emotions that news content evokes influences users' engagement with news content (Gupta & Yang, 2019; Kumar et al., 2018). Users are more likely to disseminate or comment on news posts with positive emotions. On the other hand, studies have demonstrated that news content, which arouses negative emotions tend to obtain more retweets than non-news posts (Hansen et al., 2011). Users tend to express negative or positive emotions (resonance and dissonance) in their reactions to the news post. The fact publishers use different emotional language in their social media posts, which in turn influences user expression or reaction (Aldous et al., 2021). The emotions in user comments, although they are currently underutilized, have the potential to improve fake news detection algorithms.

The flow of affective information is studied by Ghanem et al. (2021) to evaluate the credibility of fake news in long articles. Similarly, plug-in based on dual emotion (publisher and social emotions) is proposed to improve misinformation detection systems (Zhang et al., 2021b). Studies that take a propagation-based approach integrate socially conflicting perspectives on the event by building a network of affinities within it (Jin et al., 2016). While the methods described above successfully integrate emotions into fake news detection systems, they are limited to news content and fail to incorporate emotions elicited in comments. Therefore, extracting enriched dual emotion features, present in news content and user comments can increase the accuracy of fake news detectors.

The authors of fake information usually aim to disseminate information widely to attract attention. News content exhibits emotions different from emotions extracted from user comments. This difference is recognized via emotion resonance (the emotion is analogous to social emotion) and emotional dissonance (the emotion is distinguishable from social emotion) to news users or consumers. Emotions resonance implies that the news content may exhibit a dual emotion category (e.g., both sad and happy) feelings fashioned to identical news content by both the publisher and the user. However, emotion dissonance, on the other hand shows conflicting dual emotions in news content, for example being happy on the part of the publisher's emotions but disgust or anger to social emotions in fake news. Therefore, fake news, which has different emotional dissonance and resonance than real news, can help distinguish fake news from real news. Fig. 1 shows two representative examples selected from Twitter.² In Fig. 1(i), the publisher writes a post expressing anger (e.g., killed, evil) to prompt intense emotions. Users consequently respond with anger (e.g., killing, died), doubt (e.g., scary), and mixed expressions (e.g., good). While in Fig. 1(ii), the publisher writes a post expressing fear in the news content, and users express their feelings of danger (e.g., crisis, killing) and doubt (e.g., believe? curb) in comments. Additionally, studies have shown emotional cues from news content are pertinent in detecting fake news. In contrast to fake news that evokes surprise, disgust, and fear: legitimate news content inspires trust, joy, sadness, and anticipation (Vosoughi et al., 2018). These inferences, therefore, emphasize the significance of examining and comprehending dual emotions in news content and user comments in enriching fake news detectors.

To characterize the two emotions: publisher emotions are the emotions of the publishers present in the news content while posting on social media sites. Social emotions refer to the emotions of the crowd or users when responding to social media posts. We believe that fake news detection systems should consider the relationship between these two emotions often present in fake news. To this end, this study proposes a Deep Normalized Attention-based mechanism for enriched extraction of dual emotion features and Adaptive Genetic Weight Update-Random Forest (AGWu-RF) for classification. First, the deep normalized attention-based mechanism incorporates BiGRU that improve feature value by extracting long-range context information to eliminate gradient explosion issues. The genetic weight for the model is adjusted to RF and updated to achieve optimized hyper parameter values that support the classifiers' detection accuracy.

Our main contributions are:

- We propose a deep normalized attention-based mechanism for enriched extraction of dual emotion features. Specifically, we use stacked BiGRU to extract dual emotional features in a feature vector space. Attention mechanism is incorporated to also learn long-term dependencies and improve feature value by extracting long-range context information to eliminate gradient explosion issues.
- We propose an Adaptive Genetic Weight Update-Random Forest (AGWu-RF) to improve detection accuracy. Adaptive genetic weight modifies the RF model to achieve optimal feature selection, contributing to enhanced model accuracy.
- We evaluate the proposed model for accuracy, recall, F1 score, and precision on three benchmark datasets to verify fake news detection efficiency.

2. Related work

Fake news is closely related to most recognized terms such as misinformation, disinformation, rumor, propaganda, etc. (Jung et al., 2020). Hence, the concept is used interchangeably in these contexts. The main difference, however, is the intentions of the propagators. For example, while rumor is news pieces whose veracity is uncertain at the time of dissemination; but which could still be true,

² <https://twitter.com/>

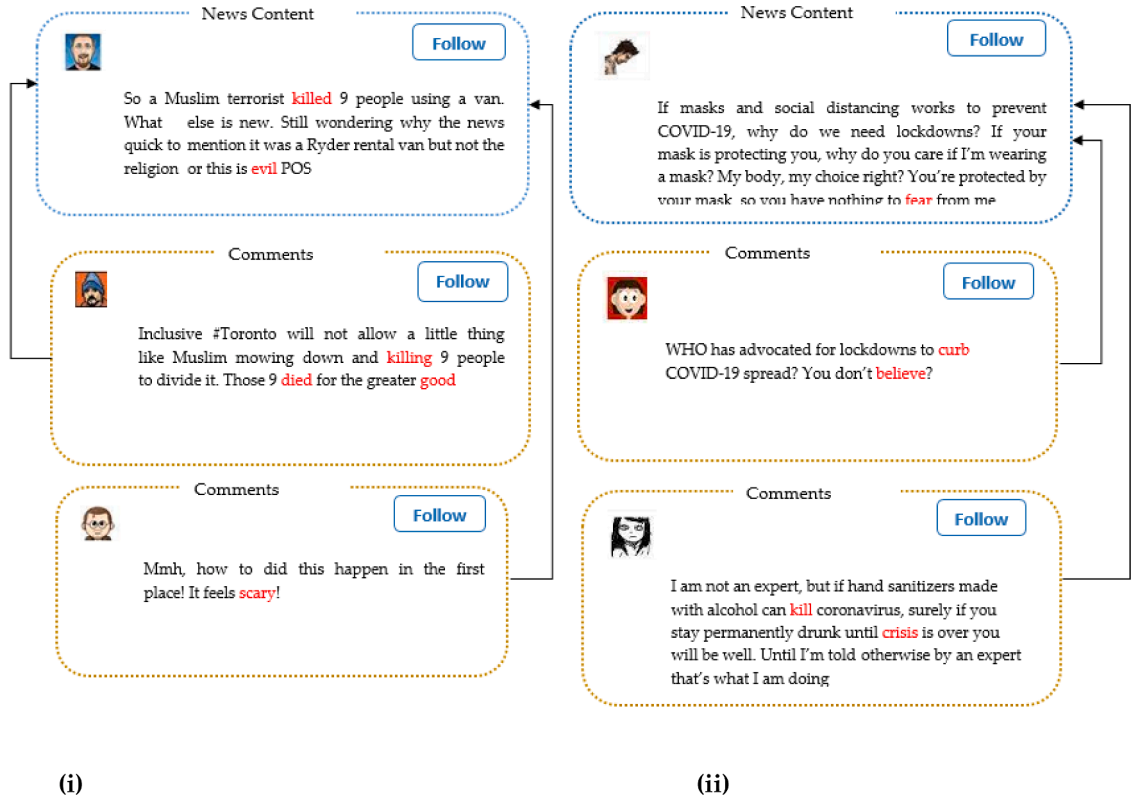


Fig. 1. Two examples of fake news in our dataset are (i) a news story with intense emotions expressed in both the content and the comments; and (ii) a piece of fake news evoking intense emotions in the narrative and prompting users to express conflicting emotions in news comments.

false, or unconfirmed (Zubiaga et al., 2018); Fake news, on the other hand, is demonstrably false and intentionally misleading news content (Shelke & Attar, 2022). In this study, we adopt the latter.

2.1. Attention-based hybrid models fake news detection

Existing works use different methods to identify fake news. The work by Nasir et al. (2021) proposes a hybrid model called Convolutional Neural Network-Recurrent Neural Network (CNN-RNN) to detect fake news in Twitter datasets. Similarly, Albahar (2021) in his work proposes a hybrid Recurrent Neural Network (RNN) and Support Vector Machine (SVM) to detect fake and real news. First, the RNN with bidirectional gated recurrent units (GRU) encodes textual data (comments and news articles) into numeric feature vectors. Then the features are passed into SVM, which uses the Radial Basis Function (RBF) for classification. Guo (2022), proposed a mutual attention (MANN) neural network that learns intermodal relationships. The model consists of a multimodal feature extractor, an attention mechanism, a fake news detector, and an event discriminator. Even though the model integrates the text and visual elements using attention, only the top features are incorporated thus limiting the model's effectiveness. In contrast, our approach uses an attention mechanism to focus on relevant dual-emotion features in a single modality. The work by Ni et al. (2021) presents a Multi-View Attention (MVAN) model for identifying fake news. The model's novelty resides in using a dual attention mechanism to capture information and signals of the original tweet and its propagation structure. Further, the attention mechanisms aid the model spot suspect users and signal phrases in the propagation structure texts. Yuan et al. (2021) suggest a Domain Adversarial and Graph Attention Neural Network (DAGA-NN) to improve false news detection system. The framework identifies fake news in all domains by supporting positive associations between two identical news node labels (effects through links between fake and fake, true and true). In addition to removing noise, the framework also blocks noise interaction between nodes (impact of false and true links) to detect fake news.

Liao et al. (2021) proposed a multi-task learning model (FDML) that examines the effect of topic tags on fake news. The model also integrates contextual evidence to increase recognition when short posts are combined. The two main characteristics of the model are linked to multitask learning and representation learning, which are applied to simultaneously train the task of categorizing news and detecting fake news. However, the model's performance is weak because author information is ignored. Also, GRUs are increasingly used to detect fake news due to their high computational efficiency. Inan (2022) proposed zoka, a model that exploits the graph attention network. The approach employs two methods to identify false news. First, it identifies possible fake users likely to spread false information. Second, the model constructs graphs with user comments, profiles, and disseminated news content. Then it filters out

possible fake users and adopts the Edge-Weighted Graph attention network for detection. Deepak and Chitturi (2020) noted that it is a challenge to resolve which of the gated RNNs (GRU and LSTM) performs better. They make a conclusion based on available computing resources. Shu et al. (2019a), proposed a sentence-comment attention model termed DEFEND. The model evaluates user comments and news items to aid fake news detection. Bi-GRU is applied to extract textual information. The model's low learning efficiency is the most notable feature. Singhanian et al. (2017), adopted a bi-GRU for word-for-word annotation. Using the preceding approach, the model captures the meaning of the words within a sentence. A framework built on top of a heterogeneous graph neural network is proposed by Li et al. (2022b). The framework learns the nodes and relationships collectively and cooperatively. The advantage of the framework is learning the fine-grained node representation utilizing a two-stage attention mechanism. Besides, Zhang et al. (2022) proposed a network dynamic GCN model that finds the symmetry between the size of the neural network and the social network. The method is practical for real-world applications such as viral marketing, fake news identification, and recommendation systems. An appropriate node label is generated for each node training using a leader-fake labeling process. Then, the selection of a group of nodes with more influence within the network is obtained using the learned node representation.

2.2. Fact-checkers

The role of fact-checkers in correcting misinformation and preventing the spread of misleading or incorrect information is undisputed. Researchers have proposed or developed ante hoc or post hoc fact-checkers to debunk misinformation on the internet. For instance, Vo and Lee (2018) designed Guardian, a fact-checking system. The Guardian cites fact-checking URLs to demonstrate people correcting inaccurate information online. The system's unique feature is cross-checking claims flagged by previous fact-checkers. Kryściński et al. (2020) proposed an unsupervised BERT-based model with weak supervision to assess factual consistency. The model considered specialized modules for performing document checks and declaring the source part of the document. The explanatory modules incorporated in the model were beneficial. They allowed human evaluation to broaden the model and confirm the document's accuracy.

Giachanou et al. (2022) focused on differentiating between users spreading false information (spreaders) and users ensuring the accuracy of the articles (checkers) and proposes Checker-Or-Spreader model. The model, which is based on CNN, included six psycholinguistic features, including emotions. Mutual Information (MI) analysis, shows that trust ranks first among emotional features; suggesting an emotional classifier can be designed to separate spreaders and checkers based on emotional characteristics. Bagade et al. (2020) introduced a fact-checking system called Kauwa-Kaate for checking fake articles. Kauwa-Kaate compares articles to trusted and crawled fact-checking sites. The main features of the system allow verification of inter-modality information (text, images, and videos). Currently, the system is limited in verifying articles unavailable on trusted and crawled fact-checking sites.

Similarly, Paschalides et al. (2021) proposed a Check-It system for detecting online fake news. The system uses a linguistic model and accepts features extracted from the news headline and body. These features are fed into a deep neural network for classification. Although the system is lightweight and easy to install as a browser plug-in, its robustness limits its effectiveness. Sengupta et al. (2021) introduced the proBlock fact-checking model. The system employs a secure voting method in which reviewers provide feedback and, using a probability measure and the feedback they have received, predict news authenticity. The model's multimodality simplifies verifying news items presented as text, video, audio, or any combination. Additionally, the system is scalable for concurrent voting and real-time feedback. Berendt et al. (2021) proposed a Fact-Rank system for claim detection in Dutch. Fact-Rank uses a deep neural architecture and expert-curated codebook to rank a claim's worthiness. First, the claim is analyzed for factuality and relevance, and when proven, the coders assign different codes: either not factual, factual, relevant, error, factual, or irrelevant. Despite its strength, the system is in the Dutch language limiting universal applicability.

Like the above studies, our approach can complement existing fact-checkers as a fact-checker extension, adding the human element in detecting fake news in short and long articles; since most automated fact-checkers rely primarily on features rather than integrating dual emotions. Also, since fact-checkers are concerned with accuracy (rather than how the results are derived), the model is efficient in increasing detection performance. However, the process is unsuitable for early detection as there are not enough comments to enable reliable detection.

2.3. Fake news detection based on emotions

Fake news spreads virally due to the emotional reaction it elicits in readers (through sharing, commenting, liking, etc.). Thus, recently, emotional features have been incorporated into false news detection (Giachanou et al., 2021; Ghanem et al., 2021; Kumari et al., 2021; Guo et al., 2019; Giachanou et al., 2019; Wu & Rao, 2020). In this regard, (Giachanou et al., 2021) proposed Long-Short Term Memory (LSTM) model to extract emotional cues from the text of the claims. The model shows the effectiveness of emotions in assessing claim authenticity. Ghanem et al. (2021) assess the affective flow of information in fake news using deep learning. Affective flows are studied by the model using a combination of extracted affective information and topics.

Using emotion as a support task for fake news detection is recommended by Kumari et al. (2021). The study uses a multi-tasking which demonstrates effectiveness in using emotional features to identify false information. Some studies propose models that simultaneously learn emotion-aware representations of comments and news content (Guo et al., 2019). Contrastingly, Giachanou et al. (2019), investigated the utility of emotional signals in the task of identifying false news by incorporating emotional cues into an LSTM model to evaluate the veracity of claims. Zarrabian and Hassani-Abharian (2020) as well as Ghanem et al. (2020) showed that while credible news contains emotion intensity and likelihood related to trust, sadness, and joy; fake news contains emotion intensity and probabilities related to surprise, disgust, and fear in their reactions. Kumari et al. (2022) proposed two models in their work; a

framework for deep learning and machine learning to identify fake news. The models leveraged the novelty and emotion-aware presentation of paired news content to detect fake news. The primary strengths of the models were the extraction of novelty and emotion-aware representations using pre-trained models; the addition of scaffold labels to the representation, and the passage of the combined representation through an LR for classification. Guo et al. (2019) suggest an Emotion-Based Fake News Framework (EFN). Through the RNN word encoder, EFN learns how to represent text features. Bi-GRU is used to learn more advanced features in text in both directions. Using the Bi-GRU, emotional features are learned in the same way as words. The most important element of the model is that gates are used.

Wu and Rao (2020) propose Adaptive Interaction Fusion Networks (AIFN) that employ Gated Adaptive Interaction Networks (GAIN) to detect semantic overlap and divergence between posts and comments automatically. The AIFN learns from input through BERT embeddings of comments and posts. Using pre-trained emotion embedding, the emotions of a word are shown in n -dimensions of that word. Projecting comparable words into neighboring spaces is the idea behind this technique, which aims to help the network learn rich word representation. The BiLSTM is then used to encode word and emotion sequences. To improve interaction, the pooled word vectors and emotion levels in posts and comments are concatenated, and GAINs (Gating Adaptive Interaction Networks) are used to capture similar semantics and different semantics adaptively. A dual emotion (publisher and social emotions) is adopted to incorporate user and publisher emotions in fake news detection systems (Zhang et al., 2021b). The features considered include Emotion Category, Emotion Lexicon, Emotion Intensity, and Sentiment Scores. These features are concatenated and combined with a fake news detector for fake news classification. In contrast with previous studies that only extract emotions from news content, the study extracts emotions from news comments, emphasizing its primary novelty in improving fake news detectors. However, the study focuses on the analysis, proposing an emotion-based plug-in to complement text-based fake news detectors. Similar to the study, we examine emotions in fake news; but provide a streamlined capability for better fake news classification by extracting publisher and user emotions; and corresponding features.

Analysis of previous studies suggests that various features extracted from fake news information play a role in the performance of fake news detection models. Methods that integrate attention mechanisms are essential in capturing crucial information within conversational structures. Consequently, methods using conventional approaches show low accuracy. Even though past research has examined several features and models, dual emotion features common on social media news posts receive little attention. This study aims to extract and capture optimal dual emotion features to improve detection accuracy.

3. Research objectives

3.1. Preliminary definition of terms and concepts

There is controversy over what exactly qualifies as an emotion in the literature. However, "emotions" are often characterized as a list: anger, disgust, sadness, joy, fear, and surprise (Cabanac & Cabanac, 2016). As a result, the emotional cues in fake news can be more intricate relatively than restricted to a single fundamental combination. Thus, emotions with a more presence in dissemination are the feelings the publisher of the news content desires recipients to experience (Chuai & Zhao, 2022). In this work, we consider the eight basic emotions (anticipation, anger, surprise, fear, sadness, joy, disgust, and trust) and two sentiments polarities (positive and negative) that many people believe are the prototypical and basic emotions (Mohammad & Turney, 2010).

Emotion features refers to the emotions extracted from the news contents (publisher's emotions) and user's comments (social emotions).

Semantic features serve as the fundamental abstract underpinning of meaning. Semantics explains how words with similar characteristics can belong to the same semantic realm or context (Al-Yahya et al., 2011). In this paper, we use the terms semantic features and semantic association alternately to refer to the contextual relationship between words with similar semantics in a sentence related to dual emotions. The intuition behind this is that semantics are crucial as the language of fake news is different; thus, claims need reinforcement for better detection.

We refer to affinity as the measure of similarity in meaningful semantic features (n-grams, words, and sentences). We believe that the semantic relationship between any two concepts, as exhibited by their shared semantic association, can be helpful in the context of identifying fake news.

3.2. Objectives

This section outlines the specific objectives for leveraging emotional features in fake news detection. The main goal of our research is to extract and capture dual emotion features to improve classifier performance by eliminating irrelevant features. We also provide a framework that improves accuracy in fake news detection by obtaining optimized hyper-parameter values through the weight update process. In summary, we define our novelty in this paper with the following research objectives:

RO 1. To examine the effectiveness of dual emotion features in the task of fake news detection. We assess the performance of the dual emotion feature model with state-of-the-art baselines.

RO 2. To extract pertinent features corresponding to dual emotions from the publisher's comments and social comments. We achieve this using the BiGRU deep normalized attention based Neural Network.

RO 3. To extract meaningful semantic associations in the context relating of dual emotions. The intuition behind this is that semantics are crucial as the language of fake news is different; thus, claims need reinforcement for better detection.

RO 4. To enhance the prediction accuracy of the fake news detection model. The study implements an Adaptive Genetic Weight Update-Random Forest (AGWu-RF) technique to improve detection accuracy.

4. Methodology

4.1. Proposed model flow

Fig. 2 shows the overall flow of the proposed framework. Several processes are involved. The RumourEval19 dataset is taken as input to pre-process the tweets. After pre-processing, we input sentence pairs (dual emotions-publisher and social emotion). The process is followed by pre-processing to eliminate data duplicates and inconsistencies to avoid a negative impact on the prediction rate of the model. In this phase, we remove special characters; since the special characters do not provide any value to textual understanding and also induce noise. We keep punctuation marks and emoticons (details in Section 5.3), however, we exclude emoji's since they are graphic symbol that describes things in the form of images. Then, we perform tokenization to divide input text into meaningful pieces termed tokens that assist in contextual understanding or developing the model for NLP. This process also supports interpreting the meaning of the input text by examining the sequence of words. We carry out vectorization for mapping phrases or words from vocabulary to respective real number vectors to be utilized later for identifying word similarities. The vectors serve as inputs into the Deep Normalized Attention-Convolutional Neural Network-Bidirectional Gated Recurrent Unit for extracting only relevant features that correspond to dual emotions. This process uses two GRUs. After feature extraction, we split the data into train and test. Then perform classification with the proposed Adaptive Genetic Weight Update-Random Forest.

4.2. Proposed framework

Our proposed deep normalized attention based and AGWu-RF framework is shown in Fig. 3. The model consists of three modules: (i) the embedding layer (ii) the feature extraction layer (iii) the genetic weight adaptation.

4.3. Embedding layer

4.3.1. CNN

CNN is a feed-forward neural network often adopted in various NLP tasks because of its superior performance. Weight sharing and local perception of CNN reduce the parameters hence boosting the learning models' effectiveness. We adopt CNN in our embeddings since it can extract and differentiate textual elements required for the emotion categorization system. We use a one-dimensional CNN with a convolution layer and the maximum pooling layer. The CNN layer consists of 16 filters (kernel size 5) with one stride, SAME padding, and ReLU activation. The convolution layer applies the filtering to the sentence matrix. After the convolution operation, the model retrieves features corresponding to the data. Since the dimensions of the retrieved features are high, to minimize this issue and reduce the training cost, a pooling layer is integrated after the convolution layer to minimize the dimensions of the features. The resulting vectors serve as input into a bi-GRU (Table 1).

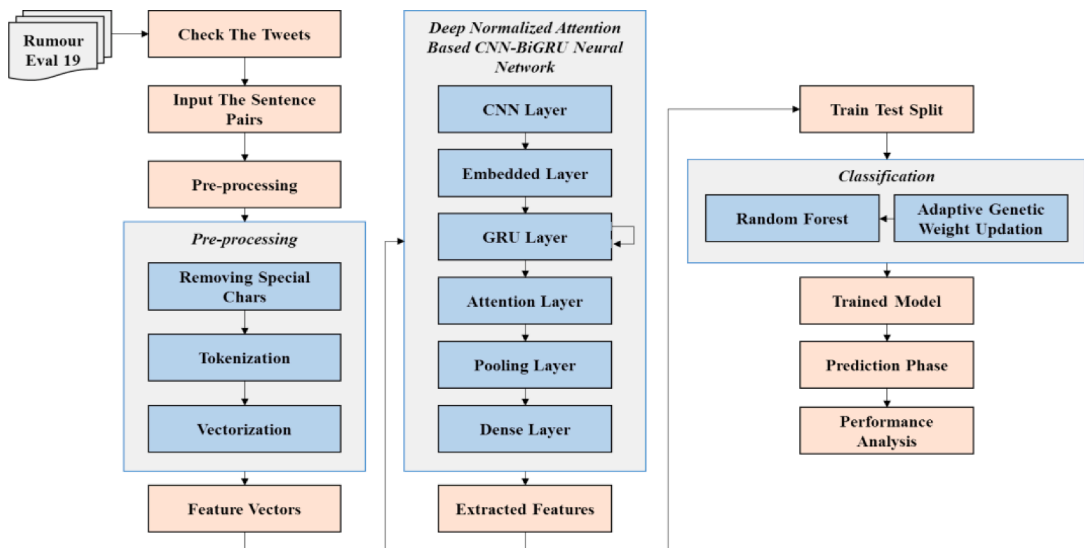


Fig. 2. The overall flow of the proposed framework.

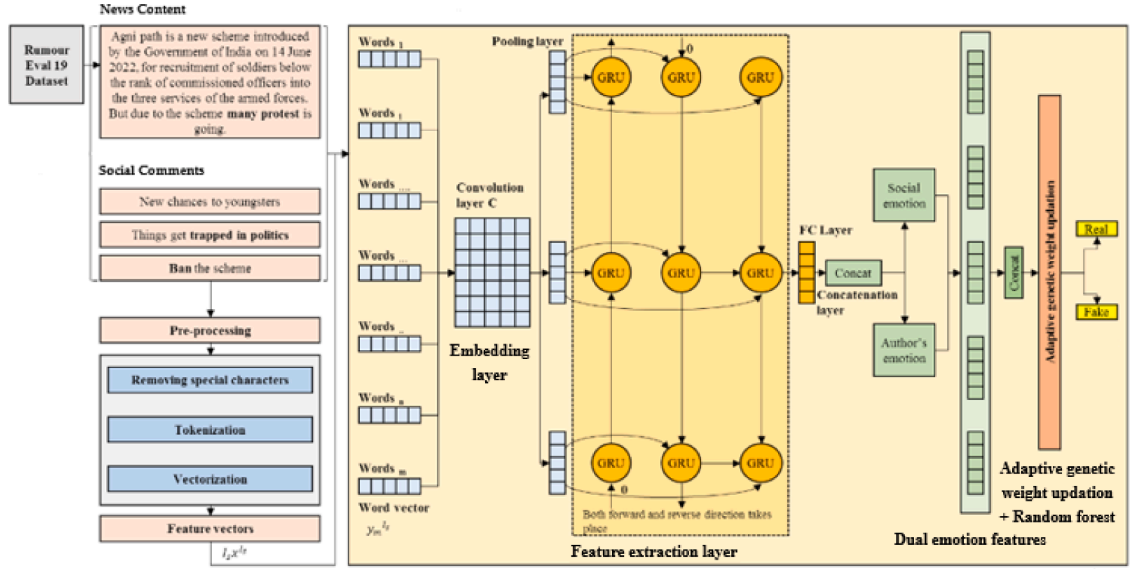


Fig. 3. Deep normalized attention based and AGWu-RF. The model has three parts: the embedding layer, the feature extraction layer, and the genetic weight adaptation. The CNN layer is an embedding layer for both the publisher comments and social comments. The stacked Bi-GRU layer extracts dual emotional features in a feature vector space during feature extraction. With the help of the attention layer, it also learns some long-term dependencies. The concatenation layer combines publishers' and social emotions and forwards to adaptive genetic weight update to obtain optimized hyper-parameter features by weight update.

Table 1

Steps entailed in the process of training the deep normalized attention model.

Model training	
Input: Sentence s , e , and labels l	
Output: Predicted label set $\hat{y}$	
1:	Initialize Bias and weights of individual deep normalized attention network
2:	for {DM not null} do
3:	$\theta f \leftarrow \theta f - t_1 (Z - \text{score}(\text{Eq.1}(y_1)))$ // Data standardization
4:	$\theta f \leftarrow \theta f - \text{CNN}(t_1 (Z - \text{score}(\text{Eq.1}(y_1))))$ // Data taken as input
5:	if $\theta f \leftarrow \theta f - \text{len}(t_1) > 1$ // Output data corresponding to the Bi-GRU layer
6:	Return
7:	$\theta f \leftarrow \theta f - \text{output}(t_1)$ // Output features are returned
8:	end for

4.4. Dual emotions feature extraction

4.4.1. Bi-GRU with attention

GRUs are streamlined and specialized networks for collecting temporal data. They are simplified version of the popular LSTM sequential learning network. GRU has two gates: the update gate and the reset gate. Similar to the LSTM network, the GRU network alters the information within the units without using individual memory cells. It is also capable of improving lengthy distances by relying on the length of words, thereby preventing over fitting on minimum samples and strengthening stability. The actual advantage of bi-GRU is that it can simultaneously extract hidden layer features in both the backward and forward orientations, which makes it effective for acquiring contextual information as shown in Fig. 5.

The input data in a GRU cell unit is standardized as shown in Eq. (1).

$$y_i = \frac{x_i - \bar{x}}{s} \quad (1)$$

where y_i indicates standardized value, x_i represents input data, $\bar{x}$ indicates the average of input data and s denotes standard deviation.

In Fig. 4, for time t , we can derive the GRU cell state as follows:

$$r_{g_t} = \sigma(w_r \cdot [h_{t-1}, x_t] + b_r) \quad (2)$$

$$z_{g_t} = \sigma(w_z \cdot [h_{t-1}, x_t] + b_z) \quad (3)$$

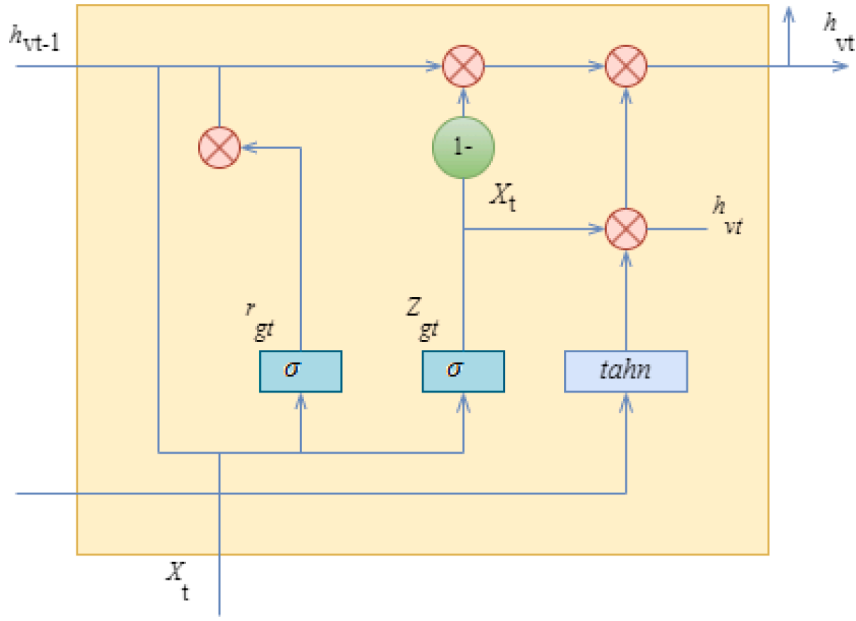


Fig. 4. The structural flow of GRU. Reset gate (r_{gt}) manages how past information subsidizes the candidate state. The update gate (z_{gt}) manages how past information is preserved and new information gets added.

$$\tilde{h}_t = \tanh(w_h \cdot [r_{gt} * h_{t-1}, x_t] + b_h) \quad (4)$$

$$h_t = (1 - z_{gt}) * h_{t-1} + z_{gt} * \tilde{h}_t \quad (5)$$

where σ is the sigmoid operation, represent dot product multiplication, b_r , b_z , and b_h represent bias parameters. x_t represent the input vector of time t . h_t represents the hidden state. It also serves as an output vector holding previous time t information. The update gate, z_{gt} , regulates the effect of the antecedent units' hidden layer output on the current state. The update gate's value influences the

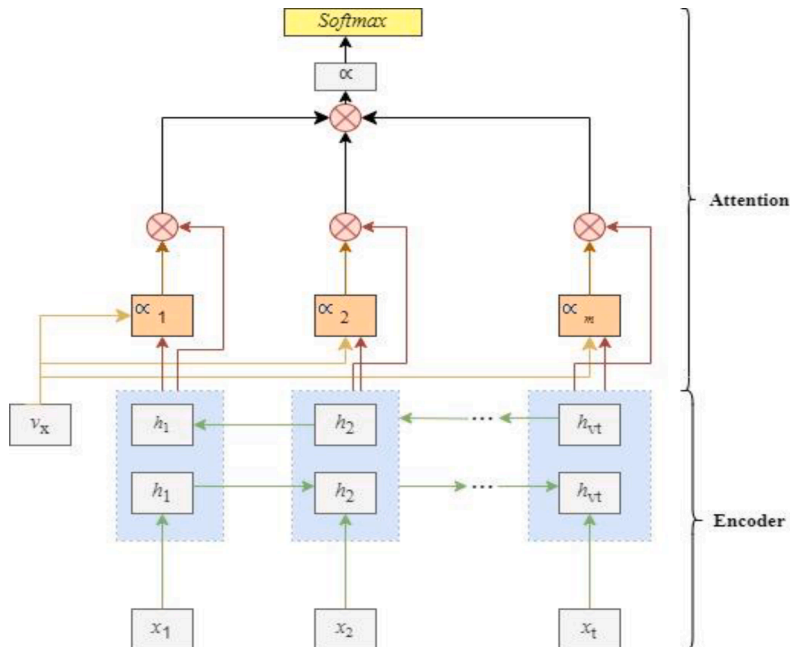


Fig. 5. The architecture of bi-GRU with an attention approach.

hidden layer output of the antecedent units. r_g is the reset gate regulate how important of h_{t-1} to $\tilde{h}_t$. The value of the reset gate determines the amount of information the antecedent unit's hidden layer disregard. $\tilde{h}_t$ determines information in the current unit that needs an update. Both gates are effective in capturing sequence length dependencies.

Our BiGRU deep normalized attention implementation uses forward GRU unit and a backward GRU unit Fig. 5). The hidden layer of the forward GRU is denoted by $\vec{h}_t$, while the hidden layer of the backward GRU unit is denoted by $\overleftarrow{h}_t$. Utilizing the Eqs. (2)–(5) and Eqs. (6)–(8), we can determine the hidden layer output of uni-directional GRU at time t . The BiGRU hidden layer output at time t is generated by combining the hidden layer of the forward and backward GRU units, as shown in Eq. (8).

$$\vec{h}_t = GRU \left(x_t, \vec{h}_{t-1} \right) \quad (6)$$

$$\overleftarrow{h}_t = GRU \left(x_t, \overleftarrow{h}_{t-1} \right) \quad (7)$$

$$h_t = \vec{h}_t, \overleftarrow{h}_t \quad (8)$$

where x_t represent the input vector of time t , and $\vec{h}_{t-1}$ and $\overleftarrow{h}_{t-1}$ is the forward and backward GRU

Since the BiGRU's goal is to capture contextual facets of the sentence: it is not more effective because the semantic information of each character in the sentence has different contributions to the dual features extraction that may cause gradient explosion and redundancy problems. We implement a deep attention mechanism in the network's second layer to improve the model and capture more important dual-emotion information (Li et al., 2022a). The approach is more capable of comprehending sequence semantics and prioritizing information. For specific text (a), it comprises of (M) words where x_m indicates m th word in document and y_m indicates m th word vector given by $m \in [1, M]$. In such an assumption, forward GRU initially reads the text (a) from (x_1) to (x_m), whereas, backward GRU accomplishes this in a reverse manner. Initially, AGWu is used to obtain v_m as a hidden indication of word annotation (h_{vt}). Subsequently, a context vector in word level (v_x) is used that is initialized randomly for computing the word significance as similarity of (v_m). Context vector (v_x) could be viewed as an information word representation at a high level and this value is updated at the training time. Lastly, significant weight is attained that is normalized by the Softmax function as shown below equations.

$$v_m = \tanh (W_x h_{vt} + b_x); \quad (7)$$

$$\alpha_m = \frac{\exp(v_m^M \cdot v_x)}{\sum_m \exp(v_m^M \cdot v_x)} \quad (8)$$

where v_x is a context word in word level, v_m represent hidden word annotation of h_{vt} , b_x is the bias and α_m represents the significant weight function.

Following this calculation, the feature vector is computed as the weighted sum of word annotations currently in decoded form and is given by Eq. (9).

$$a = \sum_m \alpha_m h_{vt} \quad (9)$$

4.5. Training

For training the model for enriched feature extraction, we normalize the input using the z-score Eq. (10). Following network initialization, the CNN and bi-GRU layers are computed. The next step, as shown in the following steps, is to calculate the output layer after classifying the attention layer.

4.6. Classification with adaptive genetic weight update – random forest (AGWU-RF)

It is important to find the ideal features for our model to provide efficient and meaningful features. We use the **AGWU-RF** to determine all the necessary features to assist in the classification and to choose a subset of features that facilitates the construction of an optimum classification model. In this instance, we consider genetic algorithms because of their inherent qualities of superior searching through complex and massive datasets. The genetic algorithm uses three steps: selection, crossover, and mutation. These techniques modify the basic chromosomal population to increase its quality. As the solution, a fitness function improves the quality of the chromosome. In these procedures, the selection process uses fitness to guide the evolution of chromosomes. Maximum fitness offers unlimited opportunities to choose suitable options. In addition, crossover refers to the incorporation of genetic material and the selection of an elite position. Individual chromosome positions are further subjected to a mutation procedure. In multiple repetitions, the successful population is determined. For optimal feature selection, we employ an adjustable inertia weight (ω) in our proposed algorithm. Adaptive inertia-weight is a subclass of a genetic algorithm that uses a two-layered structure for the chromosomes and is used in the evolution process or a discrete area; the chromosomes of successive generations are resolved directly. The location update utilizes a crossover operation, and the inertia-weight value is modified adaptively. In addition, the evolution process employs adaptive

inertia-weight for solving successive generation chromosomes directly in a discrete form. Location updating utilizes crossover and mutation operations. Unlike general operations, adaptive inertia-weight modifications increase the population and are updated by an exponential function and a global optimal value.

Let $p_i(t) = \{x_1^t, x_2^t, x_3^t, \dots, x_n^t\}$ Be a population at time (t)

Further, the inertia-weight coefficient (ω) is used to influence the velocity corresponding to a particle's varying over time and to determine the search distance between two iterations. For confirming the ability of global search and enhancing the ability of local search of GA, the inertia-weight coefficient (ω) must be enhanced adaptively. This value decides the search speed of the particle in individual iterations (Li et al., 2021). It is associated directly with the particle adaptive-value corresponding to the present iteration. Hence, the constant value (ω_0) is changed to an adaptive and dynamic form as per the below equations.

$$p_1 = \begin{cases} \omega_0 + \frac{f_i - f_{avg}}{f_{max} - f_{min}} f > f_{avg} \\ \omega_0 - \frac{f_{avg} - f_i}{f_{max} - f_{min}} f \leq f_{avg} \end{cases} \quad (11)$$

$$p_2 = \frac{t}{t_{max}} \quad (12)$$

$$\omega = e^{-p_2} + p_1 \quad (13)$$

where t_{max} represents maximum iterations, t indicates present iteration, ω_0 denotes initial value at an iteration, f_{max} indicates maximum values corresponding to adaptive inertia and f_{min} indicates minimum values corresponding to adaptive inertia and f_i denotes current fitness-value

Adaptive genetic weight modifies the RF model to achieve optimal feature selection, contributing to enhanced model performance (shown in Algorithm i). For optimized weight, several (i) trees and random vector ($T_s i$) utilizing bootstrap samples are produced independent of earlier random vectors, with a similar distribution. Further, a tree is developed using $T_s i$ and a training set. After weight update, classification is performed.

5. Datasets and experiments

In this section, we conduct experiments to evaluate the performance of our framework and other baselines in improving the performance of fake news detection.

5.1. Dataset

We use the RumorEval19 (Gorrell et al., 2019), PHEME (Zubiaga et al., 2016), and Fakeddit (Nakamura et al., 2020) datasets for experiments and analysis. The RumorEval19 dataset consists of Twitter and Reddit discussions about rumors, and the source tweet expresses a rumor or non-rumor and related comments linked to the news event. The dataset is annotated as true, false, and unverified with related ground truth. We only use true and false news labels in the data set for our experiments. The PHEME dataset is used for the binary classification of fake news in terms of tweets (news) and corresponding user comments. The dataset uses twitter threads related to nine newsworthy events; the Siege of Sydney, Charlie Hebdo, the Ottawa shooting, etc. A thread consists of a tweet (news) and a series of comments. We use the news source's original tweet and associated comments. Fakeddit is a multimodal fake news dataset. The dataset consists of one million samples of false and real information divided into six categories. It also includes real and fake news binary ground truths for classifications and fine-grain three and six-class classifications. We create training, validation, and testing subsets for PHEME and Fakeddit datasets and use Precision (P), Recall (R), and the F1 score as our evaluation metrics. The statistics of our dataset are presented in Table 1.

Algorithm-I

AGWu-RF (Adaptive genetic weight update-random forest).

		Input: GA weights $\rightarrow$ data
		Output: Class $\rightarrow$ (0/1)
1:	ω_0 is empty	
2:	for $i = 1$ to G_t // G_t : number of generated trees in a random forest // t : GA weights	
3:	$T_s i = \text{bootstrapSample}(T_s) // T_s$: training sample	
4:	$r_{tc} = \text{BuildRandomTreeClassifiers}(T_s i, t)$	
5:	$\omega_0 = \omega_0 \cup \{r_{tc}\}$	
6:	next i	
7:	return ω_0	

5.2. Performance metrics

The performance metrics considered for analysis include accuracy, recall, F1-score, and precision.

A Accuracy

It indicates the overall correct classification measure and is given by equation.14.

$$\text{Accuracy (A)} = \frac{\text{TruNeg} + \text{TruPos}}{\text{TruNeg} + \text{TruPos} + \text{FalNeg} + \text{FalPos}} \quad (14)$$

A Precision

It includes the computation of the total count of correct classification and is managed by incorrect classification. It is given by equation.15.

$$\text{Precision (P)} = \frac{\text{TruPos}}{\text{TruPos} + \text{FalPos}} \quad (15)$$

In equation.14 and equation.15, TruNeg indicates True Negative, FalNeg represents False Negative, TruPos denotes True Positive and FalPos indicates False Positive.

A Recall

It is defined as a proportion of retrieved and relevant text to the proportion of relevant text.

$$\text{Recall (R)} = \frac{\text{Relevant}_{\text{text}} \cap \text{Retrieved}_{\text{text}}}{\text{Relevant}_{\text{text}}} \quad (16)$$

A F1-score

It is also called F-measure and could be stated as the harmonic mean of P and R. It provides effective computation and is given by equation.17.

$$\text{F1 - score} = \frac{2 * (P * R)}{P + R} \quad (17)$$

5.3. Experimental setup and implementation

We cleaned the text in a sequence of steps. We evaluate the model's effectiveness in terms of accuracy, recall, precision, and F1. We use weighted accuracy for a better and more accurate comparison to comparative works. We include three additional features to fully enrich dual emotions features in news content and user comments. The first feature is the emoticons (e.g.,:(sad,: happy) since they are a universal expression of emotions. We also include punctuation marks (?, !) that express the emotions and moods people disseminate; and the frequency of sentimental terms. We train the network shown in Fig. 2 using Bi-GRU with attention and an adaptive genetic algorithm. First, we use 64 convolution layer filters with 1 kernel size for word embeddings with the RELU activation function. We set padding to the SAME in our experiments. Then 128 hidden units of Bi-GRU are applied to perform dual emotion feature extraction simultaneously in reverse and forward directions. The dual emotion features are trained simultaneously in Bi-GRU, and the generated representations are appended with fake news classification labels. The idea behind this approach is that the emotional features corresponding with a particular piece of news imply that it is fake or real because of the emotions it evokes. Then a deep attention approach is integrated on the back of the 2-layer network to aggregate feature information corresponding to the top 2 layers while minimizing the output dimension. The attention mechanism allows the model to focus on and exploit the most appropriate dual-emotion features in the news content and comments. For the classification we adapt the GA algorithm. We use the selection, cross-over, and mutation operations and apply an inertia weight to adaptively find the search distance between two consecutive iterations of our dual emotion features in our data set. We followed a few tricks to ensure training stability. Some that have helped the most are using different iterations and input iterations. We use RF for our classification because it can perform high-dimensional classification. The weights generated by GA during the training process update the RF for greater accuracy. We run experiments in a Google Colab-rendered Python environment with all the necessary libraries and a 16GB GPU.

5.4. Baselines

We make comparisons with a series of baseline fake news detection methods as follows:-

5.4.1. RumorEval 19 and pheme datasets

- **AIN**(Wu & Rao, 2020) is a cross-interaction feature integration network that is adaptive and interactive. The model employs a self-awareness technique to enhance the semantic correlation between features and a gating mechanism to detect semantic conflicts in posts and comments. It also considers emotional features.
- **SAME**(Cui et al., 2019) leverages user comments and latent sentiments on an end-to-end deep embedding network. SAME also adopts the adversarial mechanism, and a regularization loss to learn semantically meaningful spaces and annotate emotions in comments in the data sources. We adjust the code³ for a fair comparison and adopt text only information.
- **DEAN**(Zhang et al., 2021a) employs a framework based on gates to learn content and annotate emotional representation for users and publishers to identify fake news.
- **SAM**(Ajao et al., 2019), is a model, which integrates emotions via an LSTM network with a Hierarchical Attention Network (HAN) for the fake news detection task. First, the model uses topic models to identify relevant words and the Latent Semantic LSA and Latent Dirichlet Allocation (LDA) to extract emotion scores.
- **BD-DL**(Zhang & Lipani, 2019), represents the accuracy of the claims using a Bayesian model. The model time-encodes user responses using an LSTM network for fake news detection after encoding claims to be verified and generating a distribution of biases from which it samples a latent variable.
- **DTCA**(Wu et al., 2020) is decision tree-based with co-attention model that seeks evidence for explainable claim verification. The model constructs decision tree-based evidence to select high credibility comments. The goal of a co-attention-self-attention network is to allow for effective interaction with claims.
- **MTL**(Wu et al., 2019) is leverage sifted multitasking learning and a joint layer for fake news detection. The shared layer employs an attention and gating mechanism to filter and select the flow of shared features between tasks.

5.4.2. Fakeddit dataset

We adopt the following baselines for the Fakeddit Dataset

- **DeepNet** (Kaliyar et al., 2020) is a tensor factorization-based method for modeling news content and user relationships in a network. A tensor is employed to express the social context of news articles formed by a combination of the user group, news, and user information; then, a coupled tensor factorization derives the underlying representation of a news article.

EMAF(Li et al., 2021a) is a multimodal model that aligns fake news by fusing multimodal entities through a fusion network. The model uses an entity-centric cross-modal interaction that preserves semantic integrity and captures details of multimodal entities. We eliminate the multimodal alignment and the cross-modal targeting component for fair comparison. **FND-NS** (Raza & Ding, 2022) is a framework that learns useful representations for predicting fake news from news content and social context. The model employs transformer architecture. The encoder learns useful representations from fake news data, and the decoder predicts future behavior based on past observations facilitating representation learning from fake news data.

5.5. Results and analysis

5.5.1. RumorEval19 and PHEME datasets

Table 2 shows the evaluation results of our proposed method compared to the baselines. We first report results on RumorEval19 and PHEME datasets. We can observe that our proposed model outperforms other emotion based model on accuracy, precision, recall and F1 scores. Specifically, our proposed dual emotion model with genetic weight update raises AIN accuracy by 12.11% and 14.55% on RumorEval and PHEME datasets, respectively. We attribute this performance to AIN exploiting semantic conflicts in posts and comments. We implemented SAME for a fair comparison. For the implementation of this model, we excluded the multimodal features and the adversarial component, and maintained other original implementation as in the original paper. The results of SAME model on Recall on RumorEval and PHEME datasets are comparable but not better than our proposed model. It is because the SAME does not consider dual emotion for fake news detection and it is specifically designed to deal with intermodal information; therefore, emotions associated with user comments are limited.

DEAN, which also exploits dual features achieves 4% and 2.1% on RumorEval and PHEME compared to SAM, but it also introduces noise that reduces its precision, and F1 scores. SAM performs better than other emotion-related baselines, with 21.7% and 18.7% improvements on accuracy on RumorEval and PHEME datasets. We consider this performance to be related to the model utilizing emotions from news stories. However, our proposed model integrates emotions from news content and user comments, increasing the model's performance. Feature based models, BDLM-DL, MTL, and DTCA performs better than some emotion based models. However, their performance is not higher than our proposed model. We consider this performance to be related to the models integrating comments and posts in the fake news detection system, and using multi-tasking and gating mechanisms to effectively select features in the datasets. However, they are worse because our proposed model captures the semantics in posts and comments and also the emotional characteristics of both. These features are then optimized via genetic weight update contributing to higher performance

³ <https://github.com/cuilimeng/SAME>

Table 2
Statistics of the datasets.

Subset	Veracity	RumourEval19		PHEME		Fakeddit	
		Content	Comments	Content	Comments	Content	Comments
Training	Fake	79	1135	80	789	116	1000
	Real	144	1905	136	680	105	1103
	Total	223	3040	216	1469	221	2103
Validation	Fake	19	824	15	762	60	900
	Real	10	404	20	700	56	600
	Total	29	1228	35	1462	116	1500
Testing	Fake	40	689	31	671	40	533
	Real	31	805	14	400	36	615
	Total	71	1494	36	1071	76	1148

accuracy.

5.5.2. Fakeddit dataset

Similarly, as shown in Table 3, our proposed model increases the accuracy by 21.2% in Fakeddit dataset demonstrating better performance over feature based models. EMAF achieves a 6.1 percent improvement in accuracy over DeepNet. This is attributed to the pre-trained BERT model that extracts rich features and focuses on entities appearing in textual data. However, our proposed model outperforms DeepNet, EMAF, and FND-NS in accuracy, precision, recall, and F1 because the models do not incorporate dual emotions and they also do not employ optimized algorithm to select features effectively. In addition, each news content and comment embodies emotional intensity (fake news expresses negative emotions evoking more intense emotions in the audience with greater intensity). Therefore, nonetheless of the source (news content or user comments), emotions like sadness, doubt, and anger are much more severe in fake news than in real news. Thus, our mode exploits these enriched features via deep attention mechanism improving their value thereby positively influencing the results. The observations suggest the importance of incorporating effective extraction and selection of dual emotion features into the models.

In terms of dataset comparison, both the RumorEval, PHEME and Fakeddit performs better suggesting scalability and generalizability of our model. However, for the RumorEval dataset, we attribute the performance, characterized by minimal discrepancy, to differences in dataset composition for true and fake news. The rationale for this performance is discussed in Li et al. (2019), Gorrell et al. (2019), which claim there may be a preference for a particular class (fake or real) due to the small number of news posts and the low level of consensus among the annotators for the dataset. For PHEME dataset, the active moderation inherited from the platform allows more posts to be filtered, resulting in less emotional content. In the Fakeddit dataset, emotions are also likely to be expressed using caricatures, and the current emotion feature extraction method is inadequate to obtain this affective information. Perhaps an emotion-aware intermodal approach could bring further improvements.

5.6. Quantitative analysis

Across the three datasets, the model performs comparatively well on test set. In Fig. 6(a) we can observe that on RumorEval dataset, the model is able to make correct prediction for 95.93% of the time for true class (real or fake) while there are minimal instances (4.06%) when the model incorrectly identifies false class (real or fake) creating Type 1 and Type II errors. Similarly, the PHEME dataset performs comparatively better and correctly identifies correct class 96.52% of the time. However, false class is identified 3.46% times Fig. 6(b). For the Fakeddit dataset, the model correctly identifies true class 96.47% of the time. However, there are minimal instances when the model incorrectly identifies false class at 3.4% creating a classification error as shown in Fig. 6(c). We can attribute this observation to the following (i) for true class, recurring terms guide the model to perform class distinction during classification. The model learns class distinction since the model's optimized algorithm selectively obtains optimized features in the dataset and allow the model to follow different behavior of the features of related classes (ii) for false class, we think this is the case when a fake news possess subtle features making it hard for the model to correctly distinguish more discernable features to correctly classify.

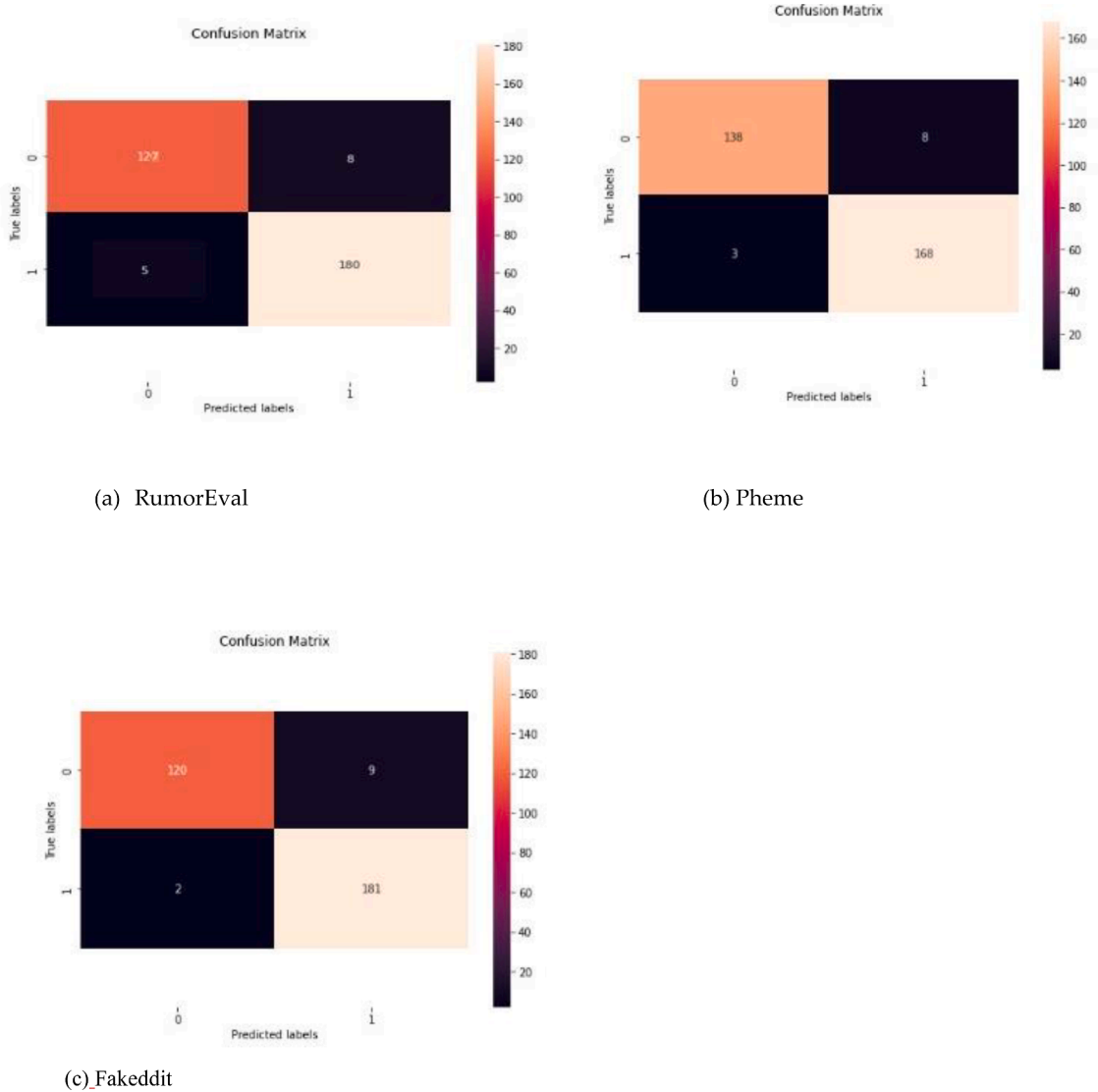
Table 3
Comparison performance on RumorEval and PHEME datasets.

Dataset	Measure	AIN	SAME	DEAN	SAM	Bayesian-DL	BDLM	DTCA	MTL	Ours
RumorEval	Accuracy (%)	82.89	0.683	0.790	0.900	80.95	80.95	82.54	81.48	0.95
	Precision (%)	78.52	0.621	0.706	0.870	77.78	77.78	78.25	72.24	0.97
	Recall (%)	86.21	0.873	0.880	0.840	77.78	77.78	85.60	86.31	0.97
	F1 score (%)	82.19	0.756	0.793	0.830	77.78	77.78	81.76	78.65	0.95
	Accuracy (%)	82.45	0.673	0.762	0.860	80.33	80.33	82.46	81.27	0.97
PHEME	Precision (%)	79.12	0.628	0.695	0.860	78.33	78.29	79.08	73.41	0.98
	Recall (%)	85.20	0.800	0.841	0.820	79.29	79.29	86.24	88.27	0.97
	F1 score (%)	82.05	0.760	0.782	0.840	78.78	78.78	82.50	80.09	0.97

Table 4

Comparison performance on Fakeddit dataset.

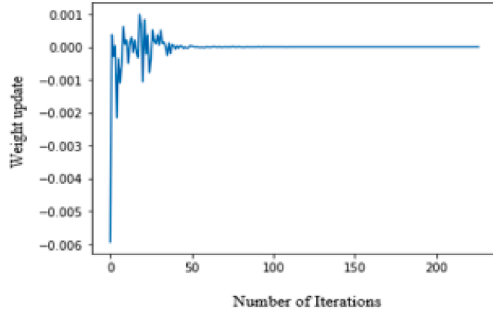
Dataset	Measure	DeepNet	EMAF	FND- NS	Ours
Fakeddit	Accuracy (%)	0.820	0.803	0.748	0.96
	Precision (%)	0.809	0.870	0.724	0.95
	Recall (%)	0.850	0.746	0.776	0.98
	F1 score (%)	0.829	0.804	0.749	0.97

**Fig. 6.** Quantitative analysis (a) RumorEval19 (b) PHEME, and (c) Fakeddit test dataset.

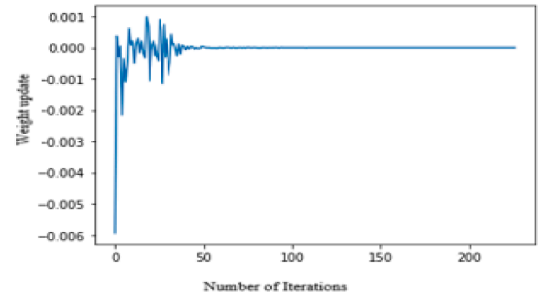
5.7. Ablation study

We analyze AGWu-RF efficacy with its variation on three datasets. The components variants are implemented as follows.

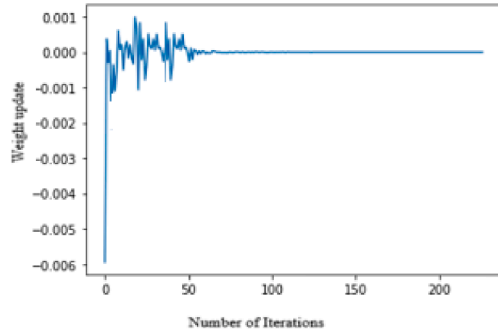
- **AGWu+RF:** We apply adaptive genetic weights to RF to enhance the possibility of making DT formation. On individual iteration, the algorithm selects varied feature combinations for theorizing optimal hyper-parameters. Generally, random forest is adopted with the inclusion of adaptive genetic weights (Fig. 7). The weight update process enhances the possibility of making DT formation (Fig. 7). The inertia weight stays fixed for every iteration. On individual iteration, the algorithm selects varied feature combinations



(a) RumorEval dataset



(b) PHEME



(c) Fakeddit

Fig. 7. Ablation analysis in terms of accuracy in three datasets. (a – c) Adaptive genetic weight update during several iterations that modifies the RF for optimal feature selection. The features are optimized in 0 – 50 iterations during which the update becomes stable in the three datasets.

for theorizing optimal hyper-parameters (Fig. 8 (a - c)). The technique utilizes different values to determine the best fit through adjustment of hyper-parameters as shown below.

$n_{\text{estimators}}$ = number of trees in the forest
 max_feature = maxnumber of features considered for spitting a node
 max_depth = maxnumber of levels in each decision tree
 min_samples_split = minnumber of data points placed in the node is split

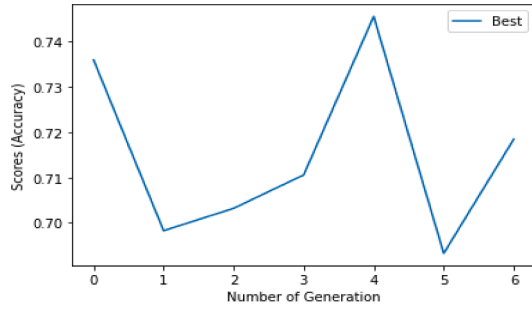
AGWu -RF: In this variation, we exclude optimization (weight update) in RF. The model is trained without including adaptive weights.

Fig. 7 shows the effectiveness of each component. The results show that AGWu+RF with optimization only outperforms AGWu-RF without weight update, demonstrating that an optimized deep normalized model select effective and enriched features corresponding to dual emotional and improves classification performance significantly. In comparison to RF, which achieves 90%, 92%, and 90% accuracy on RumorEval, PHEME and Fakeddit datasets respectively without optimization, the model achieves 95%, 97%, 96% accuracy on the same datasets. In terms of identifying fake news, the model has improved between 5% to 6% points on the three datasets.

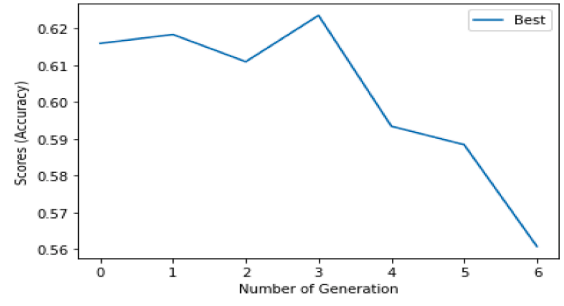
5.8. Case study analysis

To intuitively express the dual emotion traits extracted and learned from publisher content and user comments, we select a random publisher post and corresponding social comments. We then design two sub-experiments to visualize the output of BiGRU with a normalized attention mechanism. The first experiment is a complete BiGRU (BiGRU+att) with a normalized attention mechanism sub network (our feature extraction sub network), excluding the genetic weight update. The second experiment is BiGRU (BiGRU-att) without a normalized attention mechanism and the genetic weight update. We first use a max-pooling operation to pool the outputs of self-attention networks and associate them with the appropriate elements in their respective input layers.

Fig. 9(i) BiGRU with normalized attention mechanism obtains some keywords like “tense”, “situation” “gun” “shots”, etc., and also highlights the semantics associations in the news content like “multiple gunshots”. Emotion semantics links between the news content and user comments like “safe” “great” “hear” “stay” “safe” and “prayers”, etc. are also present. We draw one key observation in the



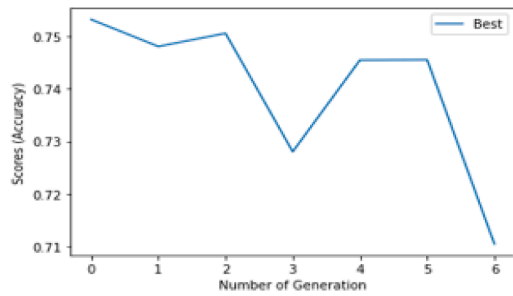
(a). RumorEval19



(b). PHEME

(a) RumorEval dataset

(b) PHEME



(c) -Fakeddit

Fig. 8. (a - c) the parameters of the algorithm adaptively vary with the number of iterations (0 - 6) as it loops through the features for optimal hyper-parameters. The hyper-parameters contribute to selecting varied feature combinations enhancing the model's accuracy in the three datasets.

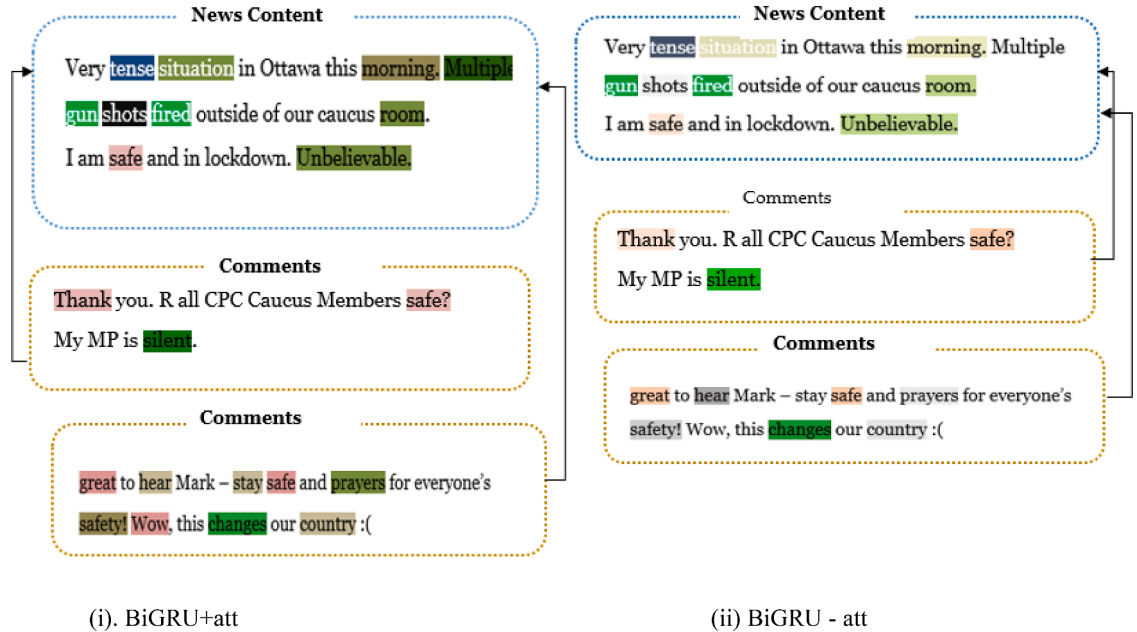


Fig. 9. Visualization of BiGRU + att and BiGRU-att.

context of “thank” and “safe”. These words appear emotionally connected because the model focuses on the context of the sentence to help learn enriched emotion highlighting the benefit of BiGRU with attention mechanism in improving dual feature correlation facilitating optimal extraction by normalized attention. In Fig. 9(ii), the model without attention mechanism also captures similar emotional features related to the publisher’s post and the user comments such as “tense”, “safe” “great” etc. However, the model does not discover keywords like “multiple” and “Wow” which would be pertinent in extractive exhaustive dual features in the news posts and user comments.

5.9. Discussion

Based on our experimental results in Tables 2 and 3, our proposed approach performs significantly better than the currently compared state-of-the-art, emotion-based, and feature-based approaches. This shows that effectively extracting and optimizing emotions in news content and user comments in a fake news detection model improve the accuracy of the detection models. Additionally, emotion-based models such as SAME and SAM incorporate publisher emotions into their models, while some feature-based models use publisher and user comments. Thus, the benefit of our system is that it incorporates dual emotions into news content, enriching the fake news detection model (Giachanou et al., 2021; Ghanem et al., 2021; Giachanou et al., 2022). For example, in the absence of user emotions; emotion-based models such as SAME and SAM report low scores of 68.3% and 90% in the RumorEval19 dataset, as well as in the precision, recall, and F1 scores, and a similar trend is reflected in the PHEME and Fakeddit datasets. Also, systems similar to our proposed model, such as AIN and DEAN, uses dual emotions and leverages conflicting viewpoints between posts and comments, and each uses a gating mechanism, achieving an accuracy of only 82.89%, 82.45%, and 12.5%, respectively; and 79%, 76.20% in the RumorEval19 data set. Both AIN and DEAN use a gating mechanism to capture emotion and enriched features, and, likely, the approach does not attend to notable features, resulting in the low performance of the models. On the other hand, the proposed model allows text to be encoded in segments into the appropriate semantic representation. The proposed framework uses word-level attention to select rich information to indicate semantic relationships. The BiGRU pays attention to features that reveal their meaning, and the convolutional layer extracts local features, semantic emotions, and morphological data corresponding to the extracted features. Similarly, emotions expressed in fake news and legitimate news can be relative. For example, anger may be higher in fake news content than in real news posts; this is comparable to the emotional expression of happiness (Zhang et al., 2021b). A similar phenomenon applies to user comments. Fake news comments are more likely to express sadness and uncertainty than credible news comments. It is highly likely that this intuition implies to the present case where our dual emotions in news content and user comments is eliciting opposing emotions influencing the overall model performance.

The higher accuracy in Table 2 and Fig. 9 (i, ii) suggests that the deep structure effectively extracts latent semantic features of the text. We could also mention that the better accuracy achieved by the model is due to the symmetry of RF’s weight update. The model uses GWU to predict dual emotion traits to achieve optimal results with fewer parameters by computing the normalized weights that minimize the regularized error during HF training. Gradient-based optimization approaches in previous work (Sheikhi, 2021) produce optimal distortions and weights. Nevertheless, the tendency of gradient-based algorithms to get trapped in local minima has prompted researchers to initiate research into meta-heuristic algorithms Li et al. (2021b). In the present study, optimal biases and weights are generated adaptively in the random forest by improving the performance of traditional GA to provide an initial population and achieve crossover and mutations.

5.10. Theoretical and practical implications

5.10.1. Theoretical implications

In practice, fake news is an evolving phenomenon that requires concerted efforts to mitigate its negative impacts. This study offers a new viewpoint by incorporating dual emotional information to spot fake news. In this context, it presents a mechanism that uses deep attention to extract dual emotion features and help eliminate the gradient explosion in the task of fake news detection. Also, the study recommends an optimization mechanism that improves the detection accuracy of fake news. It introduces a hybridized Adaptive Genetic Weight Update - Random Forest algorithm applied in the classification process. The proposed algorithm is adapted to RF and updated to give the model optimized hyper parameter values that help improve detection accuracy, hence; the study demonstrates the efficiency of dual emotion features and adaptive algorithms in improving fake news detection tasks.

5.10.2. Practical implications

The following are the implications of the study for practice: First, the study emphasizes the role of fake news management. It is tedious and time-consuming since it requires an accurate distinction between truthful and fabricated information. This research validates the importance of dual emotion features in the fight against misinformation since emotion serves as an agent in information diffusion (Li et al., 2022a). It suggests that efficient feature extraction is crucial in evaluating news veracity. The proposed solution can work with text-based fact-checking tools in a real-world setting since it is simple to prevent the spread of fake news (Allen et al., 2021). In addition, the model provides a more efficient method to modify and apply features and classifiers to select specific features to advance the study of fake news. Thirdly, we believe that research focusing on feature selection and classifiers could significantly improve the accuracy of misinformation detection, as this phenomenon affects the model’s efficiency. Finally, our methodology can serve as a solid foundation for future research. The hybridization strategy we selected is generalizable and could be applied to related tasks, such as multimodal fake news.

6. Conclusion and future works

Although several techniques attempt to identify fake news, they only consider textual features and ignore dual emotion features. In this paper, we presented an efficient method to identify fake news using dual emotion features. We adopted a stacked bi-GRU layer to extract emotional features in a feature vector space. We implemented a deep attention mechanism on top of the BiGRU to improve the model and capture more important dual-emotion information. These features serve as input into the Adaptive Genetic Weight Update Random Forest, which increases the detection accuracy of the classifier by iteratively choosing the best dual features. We performed extensive experiments on three publicly available datasets. The experimental results show that the proposed model outperforms existing models in detecting fake news.

We believe our model is suitable as a text-based fact-checker extension because fact-checkers are concerned with accuracy. However, it is unsuitable for early detection; as there are not enough comments to allow reliable detection.

There are a few potential directions for future research based on the current work. First, our proposed model does not identify only expressive emotions (anticipation, anger, surprise, fear, sadness, joy, disgust, and trust); it also recognizes polarities (negative and positive) in news content and user comments. It is viable to extend our work to examine whether a specific emotion, such as trust or anger, and specific sentiment impacts the model's overall accuracy. Second, as the study aimed to extract and integrate dual emotion features using deep learning models and to optimize the features to achieve higher accuracy, we will investigate the consistency and relativity of dual emotion resonance and dissonance and their effects on the effectiveness of fake news detection, lastly, we hope to introduce a novel concept to study multimodal dual emotions in fake news and apply advanced fusion techniques to improve the cross-correlation of social media feature models (news content and comments).

CRedit authorship contribution statement

Alex Munyole Luvembe: Conceptualization, Methodology, Software. **Weimin Li:** Supervision. **Shaohua Li:** Visualization, Investigation. **Fangfang Liu:** Data curation, Writing – original draft. **Guiqiong Xu:** Writing – review & editing.

Declaration of Competing Interest

None.

Data availability

Data will be made available on request.

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